

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
13	(a)	(i)		(in a reversible reaction) the rate of forward reaction is equal to the rate of backward reaction	1			1		
		(ii)		NaOH reacts with the H ⁺ / concentration of H ⁺ decreases (1) equilibrium shifts to right and concentration of Cl ₂ decreases / concentration of HClO increases (1)			2	2		
	(b)	(i)	I	mass water = 1.62 g (1) n(CaCl ₂) = 0.0150 mol n(H ₂ O) = 0.0899 mol (1) $x = 6$ (1) award (1) only for correct answer with no working		3		3	2	3
			II	heat to constant mass	1			1		1
		(ii)		award (1) for either of following iodide ion larger than chloride ion so less attraction to sodium ion less difference in electronegativity so NaI not as ionic as NaCl	1			1		

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	(c)	(i)		permanent dipoles between molecules in ICl but Cl ₂ has only temporary dipoles / van der Waals forces between molecules (1) permanent dipoles stronger than temporary dipoles / van der Waals forces (1) or ICl larger molecule than Cl ₂ so more electrons (1) more van der Waals forces between molecules (1)	2			2		
		(ii)		octahedral	1			1		
		(iii)		[ClF ₂] ⁺ has 2 bond pairs and 2 lone pairs (1) [ClF ₂] ⁻ has 2 bond pairs and 3 lone pairs (1) student is correct - different numbers of electron pairs will give rise to different shapes (1)			3	3		
				Question 13 total	6	3	5	14	2	4